

Cosa Verrà Exploitato

- 1 EPSS contest – Anna Guccione
- 2 Cybersecurity Lab Course
- 3 11 December 2025

1. Initial Filter

```
1 df_low_cves = df[df["epss"] < 0.01]
2 df_selected_exploited = df_low_cves.dropna(subset=
  ["cve.cisaRequiredAction"])
```

- required filter (EPSS above 0.01)
- CVEs that were already reported by CISA as requiring attention (these are just 4)

2. High risk properties

```
1 df_selected_cves = df_low_cves[
2     (df_low_cves['cvss_attackComplexity'] == "LOW") &
3     (df_low_cves['cvss_privilegesRequired'] == "NONE") &
4     (df_low_cves["cvss_userInteraction"] == "NONE") &
5     (df_low_cves["cvss_confidentialityImpact"] == "HIGH") &
6     (df_low_cves["cvss_integrityImpact"] == "HIGH") &
7     (df_low_cves["cvss_availabilityImpact"] == "HIGH") &
8     (df_low_cves["cvss_baseSeverity"] == "CRITICAL") &
9     (df_low_cves["cve.cisaRequiredAction"].isnull())]
```

- CVEs that have characteristics favorable for an attack

3. Dangerous CWEs

```
1 dangerous_cwes_id = [79, 787, 89, 352, 22]
2 dangerous_cwes = []
3 for n in dangerous_cwes_id:
4     dangerous_cwes.append("CWE-" + str(n))
```

- Top 5 dangerous CWEs of 2024

3. Dangerous CWEs

```
1 def check(row):
2     for a in row['cwe_list']:
3         for b in dangerous_cwes:
4             if a == b:
5                 return True
6     return False
7 df_selected_cves = df_selected_cves[df_selected_cves.apply(check, axis=1)]
```

- New column to check if the CWE is one of those 5

4. Final results!

```
1 df_selected_cves.sort_values("epss", ascending=False).head(10)
```

	cve.id	description	cwe_list	epss	cve.cisaRequiredAction
2633	CVE-2025-59352	Dragonfly is an open source P2P-based file dis...	[CWE-22, CWE-202]	0.00563	NaN
1589	CVE-2025-58321	Delta Electronics DIALink has an Directory Tra...	[CWE-22]	0.00454	NaN
1427	CVE-2025-58447	rAthena is an open-source cross-platform massi...	[CWE-122, CWE-787]	0.00256	NaN
2452	CVE-2025-57631	SQL Injection vulnerability in TDuckCloud v.5....	[CWE-89]	0.00242	NaN
417	CVE-2025-36897	In unknown of cd_CnMsgCodecUserApi.cpp, there ...	[CWE-787]	0.00164	NaN
3086	CVE-2025-56074	A SQL Injection vulnerability was discovered i...	[CWE-89]	0.00116	NaN
1406	CVE-2025-58462	OPEXUS FOIAXpress Public Access Link (PAL) bef...	[CWE-89]	0.00099	NaN
71	CVE-2025-9791	A weakness has been identified in Tenda AC20 1...	[CWE-119, CWE-121, CWE-787]	0.00084	NaN
72	CVE-2025-9809	Out-of-bounds write in cdfs_open_cue_track in ...	[CWE-787]	0.00083	NaN
1389	CVE-2025-23343	The NVIDIA NVDebug tool contains a vulnerabili...	[CWE-22]	0.00067	NaN